What gaps exist in current lifecycle frameworks for generative AI systems, and what evidence supports the need for more comprehensive end-to-end approaches?

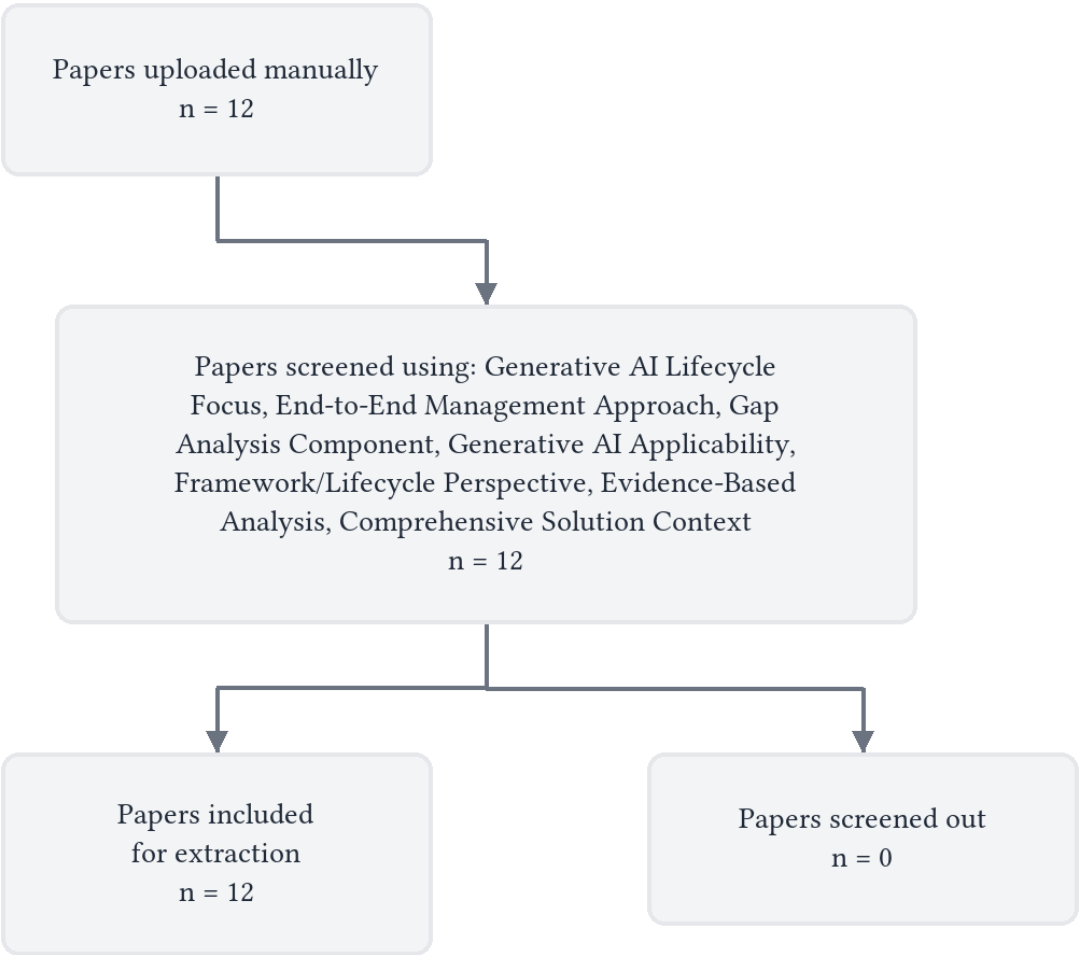
Current generative AI lifecycle frameworks suffer from fragmented stage coverage, inadequate data management and observability tooling, reactive governance mechanisms, and a fundamental architectural mismatch with the retrieval, generation, and agentic reasoning loops of modern systems — gaps whose practical consequences are evidenced by widespread deployment inefficiencies and whose resolution through integrated operational platforms has yielded measurable improvements in reliability and cost, though comprehensive empirical validation remains limited.

Abstract

Current lifecycle frameworks for generative AI systems exhibit significant gaps across multiple dimensions. Existing approaches tend to address isolated lifecycle stages rather than providing integrated end-to-end coverage [1], and traditional MLOps methods lack adequate quality control and scalability for LLMs [2]. Specific deficiencies include limited post-deployment data management support [3], absence of standardized evaluation metrics for retrieval and generation processes [3], inadequate real-time hallucination detection [2], reactive rather than proactive governance frameworks [4], and insufficient tooling for observability in RAG and agentic systems [3, 5]. The progression from MLOps to LLMOps to RAGOps to AgentOps reflects a growing recognition that each new generative AI architecture introduces lifecycle requirements — such as prompt management, retrieval pipeline monitoring, and multi-agent coordination [5] — that prior frameworks were not designed to accommodate.

Quantitative and empirical evidence supports the case for more comprehensive approaches: 55% of businesses are not producing models and 18% require over 90 days for deployment under current practices [6], while a financial services company using an integrated AgentOps platform reduced hallucination rates by 42% and cut operational costs by 35% [5]. Proposed solutions converge on continuous monitoring throughout the lifecycle [2, 7], integration of governance and ethics as first-class lifecycle stages [4, 8], and specialized tooling beyond traditional MLOps capabilities [8, 9]. However, no single framework currently addresses all dimensions — technical operations, data management, safety, governance, and agent coordination — in an integrated manner, and the evidence base remains largely theoretical or based on limited case studies rather than large-scale empirical validation.

Flow Diagram



Paper search

To answer your research question ”What gaps exist in current lifecycle frameworks for generative AI systems, and what evidence supports the need for more comprehensive end-to-end approaches?”, we looked over only the sources you uploaded.

12 sources were uploaded to Elicit for screening.

Screening

We screened in sources based on their abstracts that met these criteria:

- **Generative AI Lifecycle Focus**: Does this study examine lifecycle frameworks, methodologies, or governance approaches specifically for generative AI systems?
- **End-to-End Management Approach**: Does this research focus on end-to-end management approaches for AI systems with generative capabilities?
- **Gap Analysis Component**: Does this study identify, evaluate, or critique gaps in current AI lifecycle management practices?
- **Generative AI Applicability**: Does this study focus on generative AI systems OR does it address non-generative AI systems with clear applicability to generative AI?
- **Framework/Lifecycle Perspective**: Does this research go beyond purely technical implementation details to include framework or lifecycle considerations?
- **Evidence-Based Analysis**: Is this study more than just an opinion piece, editorial, or commentary - does it include empirical evidence or systematic analysis?
- **Comprehensive Solution Context**: Does this study address solutions within a lifecycle context rather than focusing exclusively on single-point solutions without broader framework consideration?

We considered all screening questions together and made a holistic judgement about whether to screen in each paper.

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **Lifecycle Framework**:

  Extract details about the generative AI lifecycle framework(s) described or analyzed in the study, including:

- Name/type of framework (e.g., LLMOps, RAGOps, MLOps, governance framework)
- Specific stages/phases included in the framework
- Scope of coverage (data, training, deployment, monitoring, governance, etc.)
- Whether it claims to be comprehensive or acknowledges limitations
- Target generative AI systems (LLMs, RAG, multimodal, etc.)

- **Identified Gaps**:

  Extract all gaps, limitations, or shortcomings identified in current generative AI lifecycle approaches, including:

- Specific gaps in lifecycle coverage (missing stages, incomplete processes)
- Technical limitations (monitoring, evaluation, data management)
- Governance/regulatory gaps
- Integration or coordination issues between lifecycle stages
- Scalability or operationalization challenges
- Quote exact language used to describe gaps where possible

- **Evidence for Comprehensive Approaches**:

  Extract evidence, arguments, or data that supports the need for more comprehensive end-to-end generative AI lifecycle management, including:

- Problems or failures attributed to fragmented approaches
- Quantitative evidence (statistics, study results, performance metrics)
- Case studies or examples demonstrating inadequacy of current approaches
- Expert opinions or industry reports cited as evidence
- Theoretical arguments for why comprehensive approaches are necessary

- **Lifecycle Coverage Analysis**:

  Extract analysis of what lifecycle stages are covered vs. missing in current approaches for generative AI systems, specifically noting:

- Stages that are well-addressed in current frameworks
- Stages that are poorly addressed or completely missing
- Transitions between stages that are problematic
- End-to-end integration challenges mentioned
- Focus on generative AI-specific stages (not just generic ML lifecycle)

- **Proposed Solutions**:

  Extract specific recommendations, proposals, or solutions for creating more comprehensive generative AI lifecycle frameworks, including:

- Specific methodologies or approaches proposed (e.g., full-stack safety, integrated RAGOps)
- New stages or processes that should be added
- Integration mechanisms between lifecycle phases
- Governance or regulatory recommendations
- Technical solutions for end-to-end management
- Implementation strategies or roadmaps provided

- **Study Context**:

  Extract contextual information relevant to generative AI lifecycle framework analysis, including:

- Type of generative AI systems focused on (LLMs, RAG, multimodal, etc.)
- Application domains or use cases examined
- Industry vs. academic perspective
- Methodological approach (survey, case study, theoretical, empirical)
- Scope of literature reviewed or data analyzed
- Publication year and any temporal context about framework evolution

Results

Characteristics of Included Studies

Study	Full Text Retrieved?	Study Type	Framework Focus	Target Systems	Perspective	Methodological Approach
Kun Wang et al., 2025	Yes	Survey	Full-stack safety framework [1]	LLMs [1]	Academic and industry [1]	Literature survey of 900+ papers [1]
Richard Shan et al., 2024	Yes	Framework proposal with case studies	LLMOps (4D: Discover, Distill, Deploy, Deliver) [10]	LLMs [10]	Industry-oriented [10]	Empirical examples and case studies [10]
Xiwei Xu et al., 2025	Yes	Framework proposal	RAGOps extending LLMOps [3]	RAG systems [3]	Academic [3]	Theoretical and conceptual [3]
Satyadhar Josh et al., 2025	Yes	Systematic review	Integrated MLOps, LLMOps, and AgentOps [5]	LLMs and autonomous AI agents [5]	Academic [5]	Systematic review of 100+ articles [5]
H.O. Поляковська et al., 2025	Yes	Framework proposal	LLMOps operational framework [2]	LLMs [2]	Academic [2]	Theoretical and empirical [2]
Anas Bodor et al., 2025	Yes	Case study	LLMOps with continuous monitoring [7]	LLMs [7]	Academic [7]	Case study (real estate domain) [7]
Araz Taeihagh et al., 2025	Yes	Governance analysis	Comprehensive governance framework [4]	Broad generative AI (LLMs, GANs, VAEs, diffusion models) [4]	Policy and industry [4]	Theoretical and policy analysis [4]
Josu Diaz-De-Arcaya et al., 2024	Yes	Literature review	LLMOps definition and lifecycle [8]	LLMs [8]	Academic and industry [8]	Review of scientific and grey literature [8]
Megha Sinha et al., 2024	Yes	Framework proposal	LLMOps (subset of FMOps) [9]	LLMs [9]	Industry-oriented [9]	Theoretical, based on practitioner experience [9]
Dasaprakash Krishna-murthy et al., 2025	Yes	Framework proposal	LLMOps for intelligent automation [11]	LLMs, RAG, Agentic AI [11]	Industry [11]	Theoretical [11]

Study	Full Text Retrieved?	Study Type	Framework Focus	Target Systems	Perspective	Methodological Approach
Chansung Park et al., 2024	Yes	Empirical study	LlamaDuo LLMOps pipeline [12]	LLMs [12]	Academic [12]	Empirical experiments [12]
Jasper Stone et al., 2025	Yes	Framework proposal	Unified MLOps lifecycle incorporating LLMOps [6]	LLMs [6]	Academic [6]	Theoretical [6]

All 12 sources had full texts available. The studies span 2024–2025, reflecting the recency of the LLMOps field. The majority focus on LLMs specifically, though three extend scope to RAG systems [3], autonomous agents [5], or broader generative AI modalities including GANs, VAEs, and diffusion models [4]. The methodological approaches range from large-scale literature surveys [1, 5] and theoretical framework proposals [2, 6, 9, 11] to empirical studies [12] and domain-specific case studies [7]. Notably, the mix of industry-oriented [9–11] and academic [2, 3, 5] perspectives provides complementary views on lifecycle gaps.

Thematic Analysis

Theme 1: Fragmentation Across Lifecycle Stages

A central finding across the literature is that existing frameworks tend to address isolated lifecycle stages rather than providing integrated end-to-end coverage. Kun Wang et al. (2025) observe that current surveys on LLM safety "focus on specific stages rather than the entire lifecycle, lacking a comprehensive understanding of the entire 'lifechain' of LLMs" [1]. This fragmentation is echoed by Diaz-De-Arcaya et al. (2024), who note that traditional methodologies like DevOps and MLOps are insufficient for addressing the specific challenges of LLMs [8], and that academic literature lacks comprehensive coverage of LLMOps stages [8]. Поляковська et al. (2025) similarly establish that traditional MLOps methods do not provide adequate quality control and scalability for LLMs [2].

The specific stages that are well-addressed versus poorly addressed vary across studies but reveal consistent patterns. Deployment and monitoring stages are generally well-covered in current frameworks [1, 7, 11], as are traditional DevOps-aligned phases such as planning, development, testing, and release [3]. However, several stages remain poorly addressed:

- **Pre-training and data preparation safety**: Comprehensive data preparation, pre-training safety, and post-training stages are identified as poorly covered [1]. Data governance and quality management remain persistent challenges [3, 5].
- **Ethical compliance and governance**: Multiple studies identify ethics, fairness, and regulatory compliance as stages where current frameworks fall short [2, 4, 10].
- **Observability and monitorability**: RAG-specific observability in retrieval processes [3], real-time hallucination detection [2], and decoupling different dimensions of quality monitoring [3] are consistently flagged as inadequate.
- **Commercialization and post-deployment data management**: The final commercialization stage is identified as missing from most frameworks [1], and post-deployment data-related aspects receive limited support in current LLMOps tools [3].

Theme 2: Inadequacy of Traditional MLOps for Generative AI

A recurring argument across the corpus is that generative AI systems, particularly LLMs, introduce challenges fundamentally distinct from those addressed by traditional MLOps. Sinha et al. (2024) state that "operationalizing large language models is unlike traditional AI solutioning" [9], identifying differences in personnel and processes, model selection and adaptation, evaluation and monitoring, data privacy, and technology requirements [9]. Stone et al. (2025) highlight that there is no consensus on incorporating LLMs into existing MLOps lifecycles [6], and report that 55% of businesses are not producing models and 18% require more than 90 days for deployment [6], suggesting systemic inefficiencies.

The limitations of MLOps manifest in several LLM-specific dimensions. Evaluation is inherently more subjective and complex for generative outputs [5]. Hallucination detection and mitigation require specialized monitoring absent from standard MLOps pipelines [2, 9]. The computational intensity of LLMs creates distinct scalability and cost management challenges [6, 8]. Prompt engineering, vector databases, and retrieval-augmented generation introduce entirely new lifecycle stages not contemplated in MLOps [3, 5, 9].

Josh et al. (2025) frame this as an evolutionary progression from MLOps to LLMOps to GenAIOps to AgentOps, where each successive paradigm addresses gaps left by its predecessor [5]. The transition from one operational paradigm to the next is itself problematic, requiring integration of new principles rather than simple extension [5].

Theme 3: Governance, Regulatory, and Ethical Gaps

Governance gaps feature prominently across both technical and policy-oriented studies. Taeihagh et al. (2025) provide the most extensive treatment, identifying governance challenges including data governance deficiencies, intellectual property concerns, bias amplification, privacy violations, misinformation risks, and limited public engagement in governance processes [4]. They observe that current regulatory frameworks are often reactive rather than proactive, leading to regulatory gaps that cannot keep pace with technological advancement [4]. The general-purpose nature of generative AI makes it difficult to apply regulations consistently across sectors [4].

From the technical side, Shan et al. (2024) note that "adhering to evolving regulations regarding AI and data usage is a challenge for enterprises using LLMs" [10]. Diaz-De-Arcaya et al. (2024) identify data privacy, security, and ethics and fairness as distinct LLMOps stages that require dedicated attention [8]. Xu et al. (2025) specify that RAGOps must align with regulations like GDPR and the EU AI Act [3], while the verification phase of knowledge base maintenance currently lacks mechanisms to detect malicious data insertion [3]. Sinha et al. (2024) acknowledge that GenAI frameworks are a work in progress and should evolve to accommodate ethical and social implications [9].

The security dimension is particularly acute for agent-based systems. Kun Wang et al. (2025) emphasize that multi-agent systems introduce complex attack surfaces requiring attention to agent brain security, tool invocation safety, memory retrieval protection, and communication protocol security [1]. Josh et al. (2025) extend this to include safety guardrails and session tracking as essential components of AgentOps [5].

Theme 4: Data Management as a Cross-Cutting Gap

Data management emerges as a cross-cutting deficiency that affects multiple lifecycle stages. Xu et al. (2025) observe that current LLMOps tools are primarily focused on model and prompt management, with limited support for data-related aspects post-deployment [3]. This gap motivated the development of RAGOps, which incorporates a dedicated data management lifecycle encompassing ingestion, verification, storage, and updates [3].

Krishnamurthy et al. (2025) identify data management challenges including collecting high-quality datasets and ensuring efficient storage [11]. Park et al. (2024) highlight that synthetic datasets generated by service LLMs may introduce biases and safety issues [12], and that the quality and diversity of synthetic data directly affects knowledge transfer to smaller models [12]. Bodor et al. (2025) demonstrate the importance of continuous data monitoring through a real estate case study, integrating real-time web scraping with tools like Kafka and Spark Streaming for data acquisition and processing [7].

The absence of standardized evaluation metrics across retrieval resources, retrieval processes, and generation is identified as a significant challenge that hinders accurate assessment of system performance [3]. This lack of uniformity is compounded by the stochastic nature of RAG systems, which complicates testing and reproducibility [3].

Theme 5: Integration and Scalability Challenges

End-to-end integration remains a persistent challenge. Shan et al. (2024) identify that "one of the primary obstacles lies in scaling LLMs to suit enterprise-level demands while seamlessly integrating them with existing IT infrastructure" [10]. Josh et al. (2025) propose a unified control plane that integrates MLOps, LLMOps, and AgentOps principles [5], implicitly acknowledging that current approaches lack such integration. Stone et al. (2025) report a projected 43% growth in the MLOps market [6], yet note persistent confusion from differing lifecycle frameworks and maturity models [6].

Scalability challenges manifest at multiple levels: computational resource requirements for training and inference [2, 8], the need for distributed computing methods and resource optimization [2], and the difficulty of maintaining performance as business needs grow [10]. Park et al. (2024) address this through a migration pipeline from large service LLMs to smaller local models, though the iterative fine-tuning process itself is computationally intensive [12]. The assumption of continued access to service LLMs for data generation may not always be feasible due to proprietary restrictions [12].

Taeihagh et al. (2025) add a governance dimension to scalability, noting that the public sector lacks sufficient resources and talent to effectively oversee AI developments [4], and that cross-border coordination mechanisms are needed to address global risks and avoid regulatory fragmentation [4].

Evidence Supporting the Need for Comprehensive Approaches

Evidence Type	Key Finding	Source(s)
Quantitative: market growth	43% projected growth in the MLOps market [6]	Stone et al., 2025
Quantitative: deployment inefficiency	55% of businesses not producing models; 18% requiring 90+ days for deployment [6]	Stone et al., 2025
Quantitative: operational improvement	Financial services company reduced hallucination rates by 42%, improved agent uptime to 99.97%, and cut operational costs by 35% using AgentOps [5]	Josh et al., 2025
Qualitative: MLOps inadequacy	Traditional MLOps methods do not provide adequate quality control and scalability for LLMs [2]	Поляковська et al., 2025

Evidence Type	Key Finding	Source(s)
Qualitative: lifecycle fragmentation	Current surveys focus on specific stages rather than the entire lifecycle [1]	Kun Wang et al., 2025
Qualitative: data management gaps	LLMOps tools focus mainly on model and prompt management with limited data support [3]	Xiwei Xu et al., 2025
Qualitative: governance gaps	Regulatory frameworks are often reactive rather than proactive [4]	Taeihagh et al., 2025
Empirical: pipeline validation	Extensive experiments demonstrate effectiveness of migration from service LLMs to local models [12]	Park et al., 2024
Case study: continuous monitoring	Real estate domain demonstrates benefits of integrated LLMOps and MLOps with continuous monitoring [7]	Bodor et al., 2025
Expert opinion: framework necessity	LLMOps is necessary due to fundamental differences from classic ML and MLOps [9]	Sinha et al., 2024

The quantitative evidence, while limited, is instructive. The deployment inefficiencies reported by Stone et al. (2025) — with over half of businesses failing to produce models and nearly one-fifth requiring extended deployment timelines [6] — suggest systemic gaps in current lifecycle management. The operational improvements reported by Josh et al. (2025), including a 42% reduction in hallucination rates and 35% cost reduction through comprehensive AgentOps platforms [5], provide direct evidence that integrated approaches yield measurable benefits over fragmented ones.

Proposed Solutions

The studies collectively propose several complementary approaches to address identified gaps:

Solution Category	Specific Proposals	Source(s)
Full-stack safety	Comprehensive safety covering data preparation through commercialization, including agent brain security, tool invocation safety, memory retrieval protection [1]	Kun Wang et al., 2025
Structured lifecycle (4D)	Discover, Distill, Deploy, Deliver stages with skilled teams and cross-functional collaboration [10]	Shan et al., 2024

Solution Category	Specific Proposals	Source(s)
Data-centric operations	RAGOps extending LLMOps with dedicated data management lifecycle, CI/CD integration for both query processing and data management [3]	Xu et al., 2025
Unified control plane	Integrated framework combining MLOps, LLMOps, and AgentOps with observability, debugging, and lifecycle management [5]	Josh et al., 2025
Multi-level monitoring	Distributed computing, post-processing verification, simulation stress testing, automated algorithms to reduce false positives [2]	Поляковська et al., 2025
Continuous monitoring with streaming	MLflow with Kafka and Spark Streaming for real-time data capture and model performance tracking [7]	Bodor et al., 2025
Adaptive governance	Regulatory sandboxes, novel data-sharing frameworks, red teaming, international cooperation [4]	Taeihagh et al., 2025
LLMOps standardization	Unified definition and stage taxonomy, AI-assisted prompt engineering, new tooling for LLMOps stages [8]	Diaz-De-Arcaya et al., 2024
Practitioner-oriented framework	LLMOps as FMOps subset with defined project goals, scalable architectures, vector databases, deployment strategies [9]	Sinha et al., 2024
Automation integration	LLMOps for intelligent automation with real-time monitoring, error detection, cost optimization [11]	Krishnamurthy et al., 2025
Model migration pipeline	LlamaDuo for migrating service LLMs to smaller local models via synthetic data and multi-turn fine-tuning [12]	Park et al., 2024
Unified lifecycle with maturity model	MLOps framework incorporating LLMOps with maturity evaluation, CI/CD pipelines, responsible AI integration [6]	Stone et al., 2025

The proposed solutions show convergence on several principles: the necessity of continuous monitoring throughout the lifecycle [2, 7, 11], the importance of integrating governance and ethics as first-class lifecycle stages rather than afterthoughts [3, 4, 8], and the need for specialized tooling that goes beyond traditional MLOps capabilities

[8, 9]. There is a notable divergence in scope, however, with some proposals focusing narrowly on specific pipeline segments (e.g., model migration [12] or streaming data integration [7]) while others attempt holistic coverage across the entire lifecycle [1, 5, 10].

Synthesis

The apparent consensus across all 12 sources — that current lifecycle frameworks are insufficient for generative AI — masks meaningful differences in where the most critical gaps lie and what solutions are most promising. These differences can be attributed to the distinct vantage points of the studies.

The strongest evidence for framework inadequacy comes from studies that either conduct large-scale literature analysis or report empirical metrics. Kun Wang et al. (2025), surveying over 900 papers, establish that safety coverage is fragmented across lifecycle stages [1], while Josh et al. (2025), reviewing over 100 articles and industry practices, identify critical gaps in data governance, evaluation metrics, scalability, and security [5]. The quantitative case presented by Josh et al. — where a financial services company achieved a 42% hallucination reduction and 35% cost reduction through integrated AgentOps [5] — provides the most concrete evidence that comprehensive approaches produce superior outcomes.

The divergence between industry-oriented and academic studies is instructive. Industry-focused sources such as Shan et al. (2024) and Sinha et al. (2024) emphasize practical operationalization challenges including integration with legacy IT infrastructure [10], scalability without performance degradation [10], and the need for new organizational roles and cross-functional collaboration [9, 10]. Academic sources, by contrast, tend to identify more structural gaps, such as the absence of standardized evaluation metrics [3], the inadequacy of existing observability mechanisms for retrieval processes [3], and the need for formal lifecycle stage taxonomies [8]. Both perspectives are valid within their respective contexts: enterprises face immediate integration and scalability barriers, while the research community contends with definitional ambiguity and methodological standardization.

The governance gap identified by Taeihagh et al. (2025) represents a qualitatively different category of challenge. While technical studies propose solutions such as monitoring frameworks [2], CI/CD pipelines [3], and model migration tools [12], Taeihagh et al. argue that the general-purpose nature of generative AI makes consistent regulatory application difficult across sectors [4], and that current governance approaches are reactive rather than proactive [4]. This suggests that even technically comprehensive lifecycle frameworks will remain incomplete without parallel governance innovations, such as regulatory sandboxes and international coordination mechanisms [4].

The progression from MLOps to LLMOps to RAGOps to AgentOps reflects an expanding recognition that each new system architecture introduces lifecycle requirements unaddressed by prior frameworks. RAGOps adds data management lifecycles for external knowledge bases [3], while AgentOps introduces session tracking, tool usage monitoring, and multi-agent coordination [5]. This pattern suggests that the gap is not merely one of incomplete implementation but of fundamental architectural mismatch — frameworks designed for supervised learning pipelines cannot accommodate the retrieval, generation, and agentic reasoning loops characteristic of modern generative AI systems without substantial reconceptualization.

For organizations deploying LLMs in constrained environments with privacy requirements, Park et al. (2024) demonstrate that migration pipelines from large service models to smaller local models are feasible, though dependent on synthetic data quality and access to service LLMs [12, 12]. For enterprises deploying RAG systems at scale, the RAGOps framework provides the most detailed treatment of data management integration with operational pipelines [3]. For organizations managing multi-agent systems, the integrated MLOps-LLMOps-AgentOps framework proposed by Josh et al. (2025) offers the broadest coverage [5]. No single framework currently addresses all dimensions

— technical operations, data management, safety, governance, and agent coordination — in an integrated manner, confirming the central premise that more comprehensive end-to-end approaches are needed.

References

1. Kun Wang, Guibin Zhang, Zhenhong Zhou, et al (2025) A Comprehensive Survey in LLM(-Agent) Full Stack Safety: Data, Training and Deployment. arXivorg. https://doi.org/10.48550/ARXIV.2504.15585

2. Н.О. Поляковська (2025) Understanding and Building Operational Excellence for Large Language Models. COMPUTER-INTEGRATED TECHNOLOGIES: EDUCATION, SCIENCE, PRODUCTION. https://doi.org/10.36910/6775-2524-0560-2025-58-14

3. Xiwei Xu, Hans Weytjens, Dawen Zhang, et al (2025) RAGOps: Operating and Managing Retrieval-Augmented Generation Pipelines. arXivorg. https://doi.org/10.48550/ARXIV.2506.03401

4. Araz Taeihagh (2025) Governance of Generative AI. Policy and Society. https://doi.org/10.1093/polsoc/puaf001

5. Satyadhar Josh (2025) LLMOps, AgentOps, and MLOps for Generative AI: A Comprehensive Review. International Journal of Computer Applications Technology and Research. https://doi.org/10.7753/IJCATR1407.1001

6. Jasper Stone, Raj Patel, Farbod Ghiasi, et al (2025) Navigating MLOps: Insights into Maturity, Lifecycle, Tools, and Careers. 2025 IEEE Conference on Artificial Intelligence (CAI). https://doi.org/10.1109/CAI64502.2025.00118

7. Anas Bodor, Meriem Hnida, Najima Daoudi (2025) Integration of web scraping, fine-tuning, and data enrichment in a continuous monitoring context via large language model operations. International Journal of Electrical and Computer Engineering (IJECE). https://doi.org/10.11591/ijece.v15i1.pp1027-1037

8. Josu Diaz-De-Arcaya, Juan López-De-Armentia, Raúl Miñón, et al (2024) Large Language Model Operations (LLMOps): Definition, Challenges, and Lifecycle Management. 2024 9th International Conference on Smart and Sustainable Technologies (SpliTech). https://doi.org/10.23919/SpliTech61897.2024.10612341

9. Megha Sinha, Sreekanth Menon, Ram Sagar (2024) LLMOps: Definitions, Framework and Best Practices. 2024 International Conference on Electrical, Computer and Energy Technologies (ICECET. https://doi.org/10.1109/ICECET61485.2024.10698359

10. Richard Shan, Tony Shan (2024) Enterprise LLMOps: Advancing Large Language Models Operations Practice. 2024 IEEE Cloud Summit. https://doi.org/10.1109/Cloud-Summit61220.2024.00030

11. Dasaprakash Krishnamurthy, Vinod Neelanath (2025) Establishing a Robust LLMOps Framework for Intelligent Automation: Strategies and Best Practices. 2025 Emerging Technologies for Intelligent Systems (ETIS). https://doi.org/10.1109/ETIS64005.2025.10961869

12. Chansung Park, Juyong Jiang, Fan Wang, et al (2024) LlamaDuo: LLMOps Pipeline for Seamless Migration from Service LLMs to Small-Scale Local LLMs. Annual Meeting of the Association for Computational Linguistics. https://doi.org/10.48550/ARXIV.2408.13467